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SHAZ EJ

Monetchat / SellSell / PickPic — Consolidated Project Report

Executive Technical & Product Summary

May 2026

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1. Executive Overview

Monetchat / SellSell / PickPic is an AI-native conversational marketplace platform designed for Arabic-first commerce in the GCC region.

Unlike traditional marketplaces that depend on filters and keyword search, the platform is designed around natural language interaction, semantic retrieval, and AI-assisted commerce workflows.

Core vision:

Users interact with the marketplace the same way they interact with a human salesperson.

Examples:

- “أريد سيارة عائلية بسعر مناسب”
- “find me an iPhone under 100 KD”
- “نيسان بحالة ممتازة”
- Upload image → AI generates listing automatically

The platform combines:

- conversational AI
- vector search
- hybrid semantic retrieval
- AI-assisted listing generation
- multilingual support
- image understanding
- voice interaction
- marketplace infrastructure

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2. Current Strategic Position

Current State

The project is no longer an early prototype.

The system now includes:

- functioning marketplace backend
- AI infrastructure
- semantic search
- vector database
- Dockerized deployment
- admin management system
- seller workflows
- hybrid AI retrieval
- multilingual architecture
- Arabic conversational engine

The project has moved from:

"Can we build it?"

to:

"How do we stabilize, optimize, and scale it?"

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3. Product Vision

Core Differentiator

Traditional marketplaces:

- keyword matching
- rigid filters
- manual browsing
- poor Arabic understanding

Monetchat:

- conversational search
- AI understanding
- semantic retrieval
- contextual filtering
- AI-assisted selling

The intended experience:

"Like talking to a smart salesperson instead of using a search box."

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4. System Architecture

Frontend Layer

Technology:

- Next.js 15
- React 18
- TypeScript
- TailwindCSS
- Shadcn UI
- App Router architecture

Features:

- AI chat interface
- marketplace browsing
- seller dashboard
- admin dashboard
- product pages
- image uploads
- SSE streaming

Backend Layer

Architecture:

- full-stack Next.js backend
- API routes
- modular monolith structure

Key domains:

- authentication
- products
- search
- chat
- sellers
- admin
- analytics
- AI orchestration

Database Layer

PostgreSQL

Primary transactional database.

Core tables:

- users
- sellers
- products
- categories
- product_images
- chat_sessions
- chat_messages
- search_logs
- contact_logs
- prohibited_keywords
- notifications
- audit_logs
- refresh_tokens

Capabilities:

- RBAC
- audit tracking
- moderation
- analytics
- relational marketplace storage

Vector Search Layer

Qdrant

Used for:

- semantic search
- embeddings storage
- hybrid retrieval
- recommendation systems

Architecture:

- dense vectors
- BM25 sparse vectors
- reciprocal rank fusion (RRF)

Purpose:

- human-like search
- semantic similarity
- contextual retrieval

AI Layer

Ollama Infrastructure

Current models:

- qwen2.5:14b
- qwen2.5:7b
- llama3.1:8b
- llava
- nomic-embed-text

Roles:

Model	Purpose
qwen / llama	conversational AI
nomic-embed-text	embeddings
llava	image understanding

Capabilities:

- AI chat
- semantic understanding
- image analysis
- listing generation
- embeddings
- multilingual reasoning

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5. AI Search System

Hybrid Search Pipeline

Workflow:

User Query
↓
Intent Understanding
↓
Embedding Generation
↓
Hybrid Retrieval
├─ PostgreSQL filtering
├─ Qdrant semantic similarity
↓
RRF fusion
↓
Metadata reranking
↓
Final results

Current Search Strengths

Implemented:

- semantic search
- hybrid ranking
- multilingual embeddings
- AI query preprocessing
- structured filters
- Arabic support
- vector indexing

Current Search Weaknesses

Still pending:

- stronger intent classification
- Arabic stemming
- better reranking
- confidence thresholds
- noise suppression
- category enforcement
- conversation memory refinement

Current status:

technically intelligent, but not yet fully human-like.

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6. Arabic Conversational Engine

Major Progress Achieved

A major architecture shift was implemented:

Old system:

- AI-generated Arabic responses

New system:

- deterministic Arabic response engine

Architecture:

- AI → extracts meaning
- Arabic Engine → generates premium Arabic UX

Implemented:

- query normalization
- Gulf Arabic phrasing
- deterministic responses
- confidence wrappers
- structured Arabic UX

Result:

responses now feel native to GCC users rather than translated.

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7. Marketplace Features

Buyer Features

Implemented:

- conversational search
- semantic retrieval

- product browsing
- product detail pages
- image search
- voice search
- wishlist
- AI recommendations

Seller Features

Implemented:

- AI listing generation
- image upload
- seller dashboard
- draft generation
- listing management
- AI-assisted categorization

Flow:
Upload image
↓
AI analysis
↓
Generated:

- title
- description
- category
- price suggestion
  ↓
  seller review
  ↓
  publish

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8. Admin System

Implemented:

- admin dashboard
- KPI stats
- user management
- listing moderation
- analytics
- search logs

- moderation controls

Still pending:

- advanced moderation workflows
- suspension APIs
- moderation audit trails

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9. Mobile Architecture

Planned:

- native iOS first
- SwiftUI
- later Android via Kotlin

Architecture principle:

mobile is only an interface layer.

Backend owns:

- AI
- search
- auth
- persistence
- business logic

Mobile responsibilities:

- UI
- camera
- uploads
- API calls
- local state

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10. Infrastructure

Current Deployment

Environment:

- Windows Server

- IIS
- WSL
- Docker Compose

Containers:

- app
- postgres
- qdrant
- redis
- worker

AI:

- Ollama local inference

Current Operational Weakness

Main complexity:
Windows + WSL + Docker + Ollama bridge

Weak points:

- networking fragility
- host/container communication
- operational complexity

Recommended future:

migrate toward Linux-first deployment.

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11. Dockerization Progress

Successfully completed:

- multi-stage builds
- Prisma compatibility fixes
- Dockerized stack
- IPv4 binding fixes
- standalone builds
- container orchestration

System now:

- portable
- rebuildable
- deployment-consistent

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12. Scalability Assessment

Current Capability

Current infrastructure can realistically support:

- tens of thousands of listings
- early production traffic
- AI retrieval workloads

Long-Term Scaling Plan

Future improvements:

- separate Qdrant cluster
- managed PostgreSQL
- dedicated AI inference servers
- PM2 clustering
- CDN layer
- Redis caching
- PgBouncer
- monitoring stack

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13. Security Status

Implemented:

- JWT auth
- RBAC
- rate limiting
- Prisma ORM
- audit logs
- secure cookies
- moderation rules

Pending improvements:

- stricter CORS
- upload validation
- CSRF protection
- secrets management
- backup strategy

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14. Production Readiness

Current Estimated Completion

Area	Status
AI infrastructure	90–95%
Search backend	80–85%
Marketplace backend	80%
Admin system	70%
Mobile	planning
Production hardening	60%
UX intelligence	50–60%

Overall:

approximately 80% complete technically.

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15. Current Critical Priorities

Immediate Priorities

1. Search Intelligence

Improve:

- intent detection
- filtering
- ranking
- noise removal

2. Production Stability

Complete:

- logging
- monitoring
- retries
- caching
- backup systems

3. SSE Streaming Stability

Fix:

- stream lifecycle
- controller closing issues
- IIS buffering behavior

4. AI Search UX

Goal:

move from “search engine” to “AI assistant.”

16. Infrastructure Cost Planning

Recommended MVP Architecture

DigitalOcean:

- single Linux server

RunPod:

- Ollama inference

Estimated cost:
~$340/month

This is currently the recommended balance between:

- cost
- simplicity
- scalability

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17. Business Positioning

This is not just a marketplace.

It is evolving into:

an AI marketplace engine.

Potential applications:

- classifieds
- automotive marketplaces
- property search
- AI commerce assistants
- conversational shopping
- AI-powered local commerce

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18. Strategic Assessment

What Has Been Achieved

The hardest engineering problems are already solved:

- AI infrastructure
- embeddings
- vector search
- hybrid retrieval
- multilingual AI
- local inference
- marketplace architecture

This is a major achievement.

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19. Remaining Risk Areas

Main risks now are:

- operational complexity
- UX consistency
- search quality refinement
- production hardening
- deployment reliability

Not:

- AI capability
- architecture feasibility
- marketplace backend

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20. Final Assessment

Technical Evaluation

Component	Grade
AI architecture	A
Hybrid search	A
Database design	A
Conversational UX vision	A
Infrastructure maturity	B-
Deployment stability	B-
Production readiness	B
Mobile readiness	B-
Testing coverage	C
Operational simplicity	C+

Final Conclusion

Monetchat / SellSell / PickPic is already far beyond a typical marketplace MVP.

The system now includes:

- semantic AI search
- local LLM infrastructure
- hybrid retrieval
- AI-assisted listings
- multilingual conversational UX
- scalable marketplace foundations

The project has transitioned from:

“building the system”

to:

“refining intelligence, stability, and scalability.”

The next phase is no longer invention.

The next phase is:

- integration
- optimization
- launch
- production hardening
- user acquisition
- scaling AI search quality

Most importantly:

The platform already contains the core technical moat that differentiates it from traditional marketplaces.